AL-NEELAIN UNIVERSITY

Faculty of Computer Science and Information Technology — Graduate Studies

Comprehensive Operations & User Manual: Multi-Agent Clinical AI for Diabetes
[Operations & User Manual - Al-Neelain University

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1. Manual Overview & System Architecture

This operations manual provides step-by-step instructions for running, testing, and deploying the Multi-Agent Clinical Decision Support System for Diabetes Prediction. Evolving from the student's foundational coursework project for the prerequisite 'Machine Learning' module during the previous academic semester (preserved in directory 'Old_Project_Before_Agents'), this system was engineered for Course : COSC726 AGENTIC AI at Al-Neelain University under the academic supervision of Dr. Fakhreldeen Saeed, featuring an interactive bilingual web interface (Streamlit), modular phase verification capabilities, and automated document compilation for clinical dossiers and academic reports in Microsoft Word (.docx).

2. Launching the Interactive Web Application (Streamlit)

To launch the interactive dashboard, open a terminal in the project root directory and execute:

[Quickstart Command]
python -m streamlit run app.py

Or using the central CLI orchestrator:
python run_system.py --ui

Key Application Features & Ergonomics:
• Bidirectional Language Localization: Native support for Arabic (strict RTL typography) and English (LTR).
• Biological Sex Selection: Biological triage field (Male/Female). Selecting 'Male' automatically locks and zeros the pregnancies input to enforce biological validity.
• Multi-Provider LLM Engine: Flexible selector supporting Ollama (local private deployment),

Cohere, DeepSeek, Google Gemini, OpenAI, or the offline deterministic ADA expert engine.
• Medical Criteria Analysis Tab: Dedicated diagnostic tab providing a single unified feature importance chart with shape selector (Horizontal Bar, Vertical Columns, Pie, Donut, Radar) and session state preservation.
• Pre-configured Patient Archetypes: One-click presets for Healthy (Normal), Pre-diabetes, High Risk, and Critical Emergency cohorts.
• Dynamic Multi-Agent Execution Tracker: Real-time visual cards demonstrating intermediate agent progression and quality gate clearances.
• Sanitized Patient Word Dossier: Instant download of official clinical reports with student metadata cleanly excluded.

3. Modular Phase Verification Commands

The central CLI runner enables independent testing and quality gate auditing for each phase:

Command	Execution Scope & Target Phase
python run_system.py --phase 1	Phase 1: Physiological data imputation, calibrated ML modeling, and Gates 1.1 - 1.2.
python run_system.py --phase 2	Phase 2: Multi-agent orchestration, multi-provider LLM reasoning, and Gates 2.1 - 2.3.
python run_system.py --phase 3	Phase 3: Web interface integration, criteria tab, patient report generator, and Gates 3.1 - 3.2.
python run_system.py --phase 4	Phase 4: Automated Pytest test suite, stress testing on male edge-cases, and latency benchmarks.
python run_system.py --phase all	Executes full sequential verification across all four phases (100% pass rate).

4. Catalog of Generated Documents & Artifacts

Document Name	File Path	Description
Arabic Project Proposal	Project_Proposal/Project_Proposal_Arabic.pdf	Master's research thesis proposal in Arabic
English Project Proposal	Project_Proposal/Project_Proposal_English.pdf	Academic Master's research proposal in English
Phase 1 Bilingual Report	Phase_1_Data_Modeling_and_Validation/Phase_1_Report_Bilingual.pdf	Data engineering, ML modeling & calibration
Phase 2 Bilingual Report	Phase_2_Multi_Agent_Architecture/Phase_2_Report_Bilingual.pdf	Multi-agent architecture & multi-provider LLM clinical reasoning
Phase 3 Bilingual Report	Phase_3_Interactive_UI_and_System_Integration/Phase_3_Report_Bilingual.pdf	Interactive Streamlit UI, criteria analysis tab & report engine
Phase 4 Bilingual Report	Phase_4_Verification_Testing_and_Benchmarking/Phase_4_Report_Bilingual.pdf	Automated test suites & performance benchmarking